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BHOPAL

Fundamentals of **AI and ML**

Project Report: *Daily Activity*
Health Predictor

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INTRODUCTION

In today's lifestyle, most people are busy with studies, work, or spending time on digital devices. Because of this, we often ignore our daily habits like how much we walk, how long we sleep, or how much time we spend on our phones. Over time, these small habits can affect our overall health, but we don't always realize it.

This project, “**Daily Activity Health Predictor,**” is based on this real-life problem. The idea is to use Machine Learning to analyze daily activities and predict whether a person's lifestyle is healthy or unhealthy. The goal is not just prediction, but also to create awareness about how our everyday habits impact our health.

Problem Statement

The main problem I focused on is that people do not clearly understand whether their daily routine is healthy or not. Many times, we think we are doing fine, but in reality, we might be:

- Walking very less
- Sleeping insufficiently
- Spending too much time on screens
- Not drinking enough water
- Eating junk food regularly

All these factors together decide our health condition, but they are rarely analyzed together. There is a need for a simple system that can take these inputs and give a clear answer.

Why This Problem Matters

This problem is important because lifestyle-related health issues are increasing rapidly, especially among students and young adults. Many people don't track their habits, and even if they do, they don't know how to interpret that data.

By solving this problem:

- People can become more aware of their lifestyle
- It encourages healthier habits
- It shows how AI can be used in daily life

This makes the project both practical and meaningful.

Proposed Solution

To solve this problem, I developed a Machine Learning model that takes daily activity data as input and predicts whether the lifestyle is **healthy or unhealthy**.

The user needs to enter basic details like:

- Steps walked
- Sleep hours
- Screen time
- Water intake
- Exercise
- Junk food consumption

Based on this data, the model gives an output indicating the lifestyle category. This provides a quick and simple way for users to evaluate their habits.

Dataset Description

For this project, I created my own dataset because it was difficult to find a simple and relevant dataset online that matched the idea exactly.

The dataset includes the following features:

- **Steps** → daily walking activity
- **Sleep Hours** → amount of rest
- **Screen Time** → time spent on phone/laptop
- **Water Intake** → hydration level
- **Exercise** → whether physical activity is done
- **Junk Food** → eating habits
- **Health** → final output (Healthy or Unhealthy)

The data was created based on realistic assumptions of daily human behavior.

Approach and Methodology

To build this project, I followed these steps:

1. Data Preparation

I organized the dataset in CSV format and ensured all values were properly structured.

2. Feature Selection

All the input features were selected based on their direct impact on health.

3. Splitting the Data

The dataset was divided into training and testing sets so that the model could be evaluated properly.

4. Model Selection

I used **Logistic Regression** because:

- It is simple and easy to understand
- Suitable for classification problems
- Works well with small datasets

5. Model Training

The model was trained using the training data to learn patterns between inputs and output.

6. Testing and Evaluation

The model was tested on unseen data, and accuracy was calculated to check performance.

7. User Input System

I added a feature where users can input their own daily data and get predictions instantly.

Key Decisions

While working on the project, I made some important decisions:

- I chose a **simple model** instead of a complex one to keep the project clear and understandable
- I created a **custom dataset** to make it more relevant
- I focused on **real-life usability** instead of just technical complexity
- I kept the code simple and readable

These decisions helped in completing the project effectively within the given time.

Challenges Faced

During the project, I faced a few challenges:

- Setting up Python libraries and environment
- Interpreter mismatch issues in VS Code
- Creating a meaningful dataset manually
- Handling errors during execution
- Understanding how the model actually predicts

Initially, these issues slowed down the progress, but after debugging and learning, I was able to solve them.

Results and Observations

The model was able to successfully predict whether a lifestyle is healthy or unhealthy based on the given inputs.

Some observations:

- Higher steps, good sleep, and low screen time → Healthy
- Low activity, high screen time, junk food → Unhealthy
- The model works well for basic predictions

Even though the dataset is small, the results are meaningful and realistic.

What I Learned

This project helped me learn a lot about both technical and practical aspects:

- Basics of Machine Learning
- How to build a complete ML project from scratch
- Importance of data and features
- How to debug errors
- How AI can be applied to real-life problems

It also improved my problem-solving and logical thinking skills.

CONCLUSION

The **Daily Activity Health Predictor** shows how simple Machine Learning techniques can be used to solve everyday problems. It helps users understand their lifestyle better and encourages them to make healthier choices.

Although the model is basic, it successfully demonstrates the concept and can be improved further in the future.

Future Improvements

If given more time, I would like to improve this project by:

- Adding more data for better accuracy
- Creating a mobile or web app interface
- Giving personalized health suggestions
- Adding a health score system
- Connecting it with real fitness tracking data

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